

Econ 646 Spring 2013
Homework 1

The due date Feb 8th, at the beginning of tutorial session.

1. *Macroeconomic Data*

- (a) Replicate Table 1 of King and Rebelo (2000)
- (b) Construct the same table for the data detrended with the first difference, i.e., $\Delta \log Y_t \equiv \log Y_t - \log Y_{t-1}$ is the detrended version of output.

2. Consider the following social planner's problem. She maximizes the expected utility

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t u(C_t, N_t) \right]$$

by choosing a sequence of C_t , $N_{c,t}$, $N_{i,t}$, $I_{c,t}$, and $I_{i,t}$, subject to

$$N_t = N_{c,t} + N_{i,t} \tag{1}$$

$$C_t = A_{c,t} K_{c,t}^{\alpha} N_{c,t}^{1-\alpha} \tag{2}$$

$$I_t = I_{c,t} + I_{i,t} = A_{i,t} K_{i,t}^{\alpha} N_{i,t}^{1-\alpha} \tag{3}$$

$$K_{c,t+1} = (1 - \delta) K_{c,t} + I_{c,t} \tag{4}$$

$$K_{i,t+1} = (1 - \delta) K_{i,t} + I_{i,t} \tag{5}$$

$K_{c,0}$ and $K_{i,0}$ are given. $A_{c,t}$ and $A_{i,t}$ are exogenous productivity shocks. This is a two sector economy. There are consumption goods sector and investment goods sector, each of which has its own production function. Answer the following questions.

- (a) Derive the first order conditions.
- (b) Assume that the momentary utility function u is given by $u(C_t, N_t) = \log C_t - \frac{\eta}{1+\eta} N_t^{\frac{\eta+1}{\eta}}$ where $\eta > 0$. Notice that this is a particular form of the so-called King-Plosser-Rebelo utility. Rewrite the first order conditions using this specific utility function.
- (c) Derive the optimality condition that equates the marginal rate of substitution to the marginal product of labor, i.e., $-(\partial u / \partial N_{c,t}) / (\partial u / \partial C_t) = (\partial C_t / \partial N_{c,t})$. Prove that C_t cancels out from the condition.
- (d) Prove that in this model economy, $N_{c,t}$ and $N_{i,t}$ cannot move in the same direction; i.e., $\Delta N_{c,t} \equiv N_{c,t} - N_{c,t-1} > 0$ is always accompanied by $\Delta N_{i,t} \equiv N_{i,t} - N_{i,t-1} < 0$ and vice versa.
(HINT: Question c gives a great hint.)

- (e) Since employment figures in different sectors largely comove in the real world, this model prediction is clearly counterfactual. However, we may be able to fix this problem by adopting a different utility function. Remember that the general form of the King-Plosser-Rebelo utility function is given by

$$u(C_t, N_t) = \frac{[C_t v(1 - N_t)]^{1-\sigma} - 1}{1 - \sigma}$$

where $\sigma > 0$. Assume that v is a positive, concave function: i.e., $v(\cdot) > 0$, $v'(\cdot) > 0$, and $v''(\cdot) < 0$. Do you think $N_{c,t}$ and $N_{i,t}$ can move in the same direction if we choose an appropriate form of KPR utility? If your answer is yes, explain what form will help comovement, and under what circumstance $N_{c,t}$ and $N_{i,t}$ can move in the same direction. If your answer is no, give supporting argument why you reach such a conclusion.

(HINT: Derive the first order conditions using the general form of KPR utility. Observe similarities and differences.)